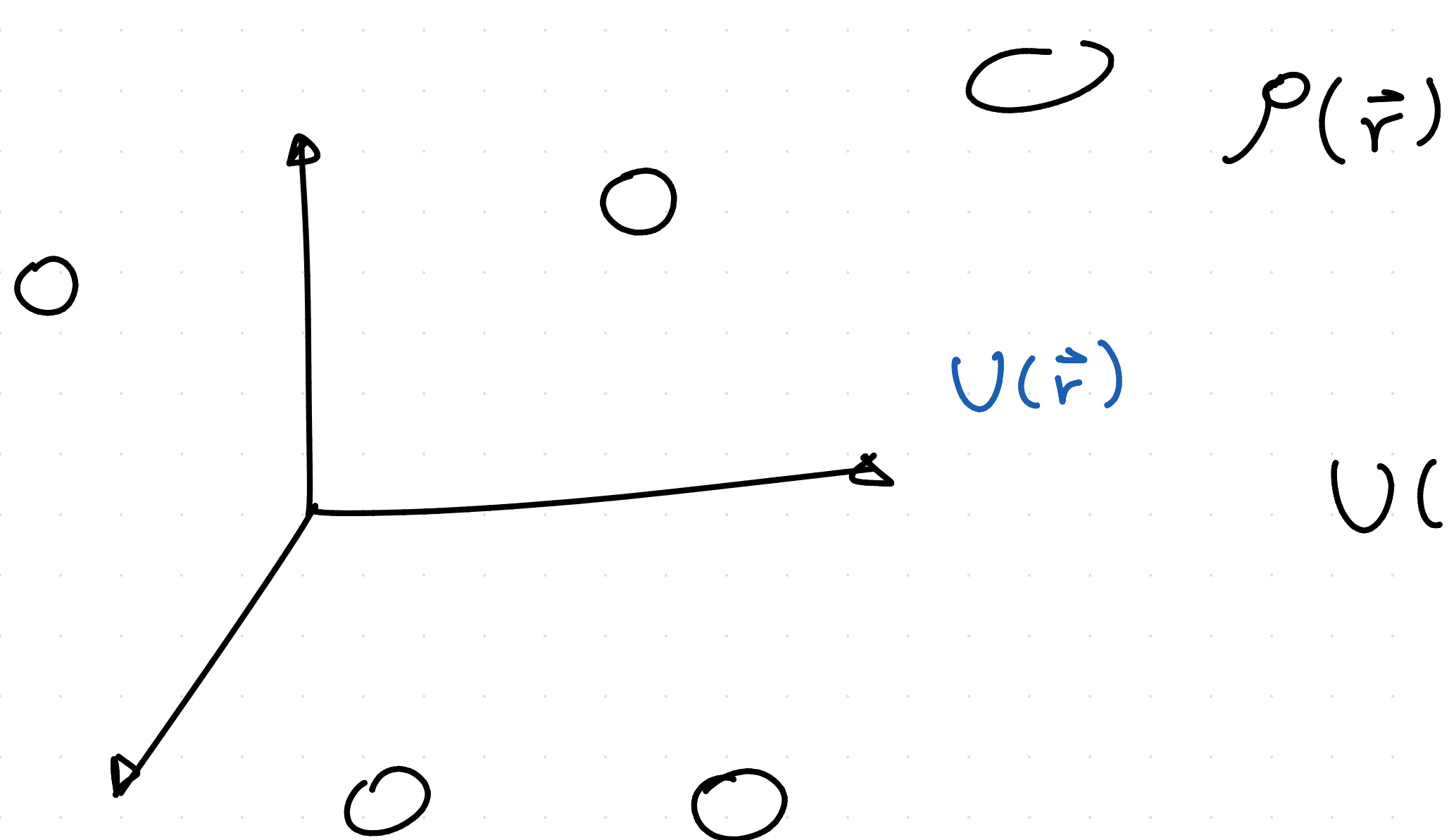
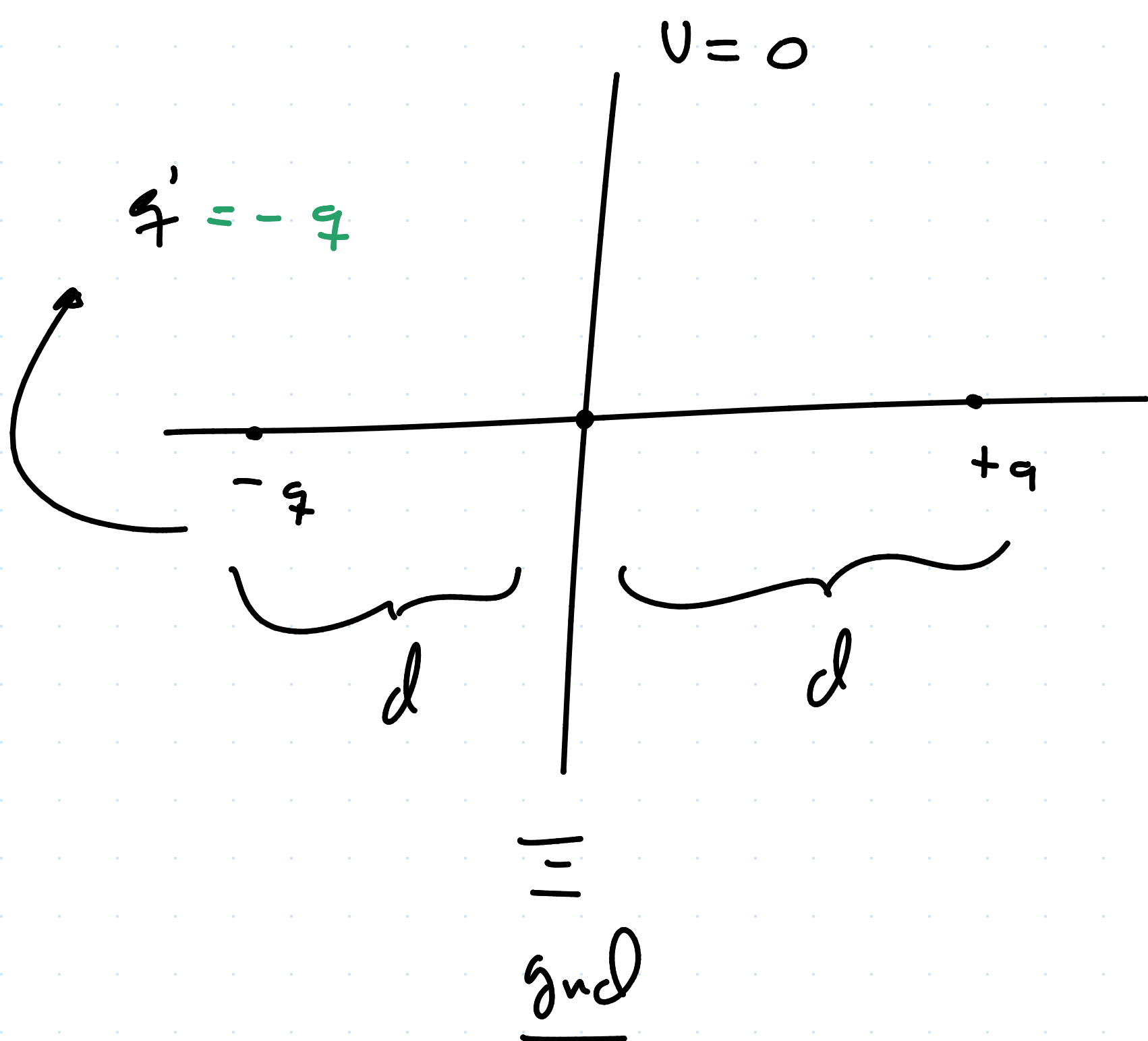


Unit 1: ElectrostaticsMethod of images

$$U(\vec{r}) = U_1 +$$

$$\frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') da'}{|\vec{r} - \vec{r}'|}$$

U_2

Example 1

$$U(\vec{r}) = U_1 + \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') da'}{|\vec{r} - \vec{r}'|}$$

$$\sigma(\vec{r}) = \epsilon_0 E_x = -\epsilon_0 \frac{\partial U}{\partial x}$$

- Tenemos

$$\nabla^2 U = 0$$

Tenemos

$$U = U_1 + \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') d\alpha'}{|\vec{r} - \vec{r}'|}$$

Tenemos

$$U_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{(x-a)^2 + (y-b)^2}}$$

Escribimos $\cup$ en forma explícita:

$$U = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{(x-a)^2 + (y-b)^2}} - \frac{1}{\sqrt{(x+a)^2 + (y-b)^2}} - \frac{1}{\sqrt{(x-a)^2 + (y+b)^2}} + \frac{1}{\sqrt{(x+(-a))^2 + (y-(-b))^2}} \right]$$

At $i) x=0 ; U=0$

$ii) y=0 ; U=0$

$$U = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{(-a)^2 + (y-b)^2}} - \frac{1}{\sqrt{(+a)^2 + (y-b)^2}} - \frac{1}{\sqrt{(-a)^2 + (y+b)^2}} + \frac{1}{\sqrt{(-(-a))^2 + (y-(-b))^2}} \right]$$

Handwritten annotations in the above equation:

- Under $(-a)^2 + (y-b)^2$: $(a^2 + (y-b)^2)^{1/2}$ (pink arrow)
- Under $(+a)^2 + (y-b)^2$: $(a^2 + (y-b)^2)^{1/2}$ (blue arrow)
- Under $(-a)^2 + (y+b)^2$: $(a^2 + (y+b)^2)^{1/2}$ (orange arrow)
- Under $(-(-a))^2 + (y-(-b))^2$: $a^2 + (y+b)^2$ (orange arrow)

Para el caso $x=0 \Rightarrow U=0$ ^ Para $y=0$ también $U=0$.

Tenemos

$$\vec{F}_q = -q \nabla U_2$$

Example 3:

$U(r, \theta, \phi) \cdot (x, y)$

$$K = q; \quad b = q; \quad q'$$

Tenemos

eliminando todo al cuadrado tenemos

$$K^2(a^2 + d^2 - 2ad \cos \theta) = a^2 + b^2 - 2ab \cos \theta$$

$$\hookrightarrow K^2 = \frac{a^2 + b^2 - 2ab \cos \theta}{a^2 + d^2 - 2ad \cos \theta}$$

~~⊗~~ real

Again!!

Tenemos

cando $r=a$

$$V(r, \theta, \phi) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{a^2 + d^2 - 2ad \cos \theta}} + \frac{q'}{\sqrt{a^2 + b^2 - 2ab \cos \theta}} \right]$$

sol. correcta

$$K^2(a^2 + d^2 - 2ad \cos \theta) = a^2 + b^2 - 2ab \cos \theta$$

$\hookrightarrow$ igualando componente a componente...

$$\Rightarrow -2adK^2 = -2ab$$

$$K^2 = \frac{b}{d}; \quad b = dK^2$$

/

$$\rightarrow \kappa^2(a^2 + d^2) = a^2 + (d\kappa^2)^2$$

$$a^2 \kappa^2 + \kappa^2 d^2 = a^2 + d^2 \kappa^4 \Rightarrow \underline{d^2 \kappa^4} + \underline{a^2} - \underline{a^2 \kappa^2} - \underline{\kappa^2 d^2} = 0$$

$$\Rightarrow d^2 \kappa^4 - \kappa^2(a^2 + d^2) + a^2 = 0$$

$$\Rightarrow \kappa^4 d^2 - \kappa^2(a^2 + d^2) + a^2 = 0$$

$$\rightarrow \kappa^2 a^2 - a^2 = d^2 \kappa^4 - \kappa^2 d^2 \Rightarrow a^2(\kappa^2 - 1) = d^2 \kappa^2(\kappa^2 - 1)$$

$$\boxed{\kappa \neq 1}$$

$$\Rightarrow d^2 \kappa^2 = a^2 \quad \} \quad \kappa = \frac{a}{d}$$

$$\hookrightarrow b = d\kappa^2 = d\left(\frac{a^2}{d^2}\right) = \frac{a^2}{d}$$

De este resultado
obtenemos...

$$b = \frac{a^2}{d} ; \quad \kappa = \frac{a}{d}$$

$$q' = [\text{por calcular}]$$

Tenemos

cálculo de q'

Partimos de:

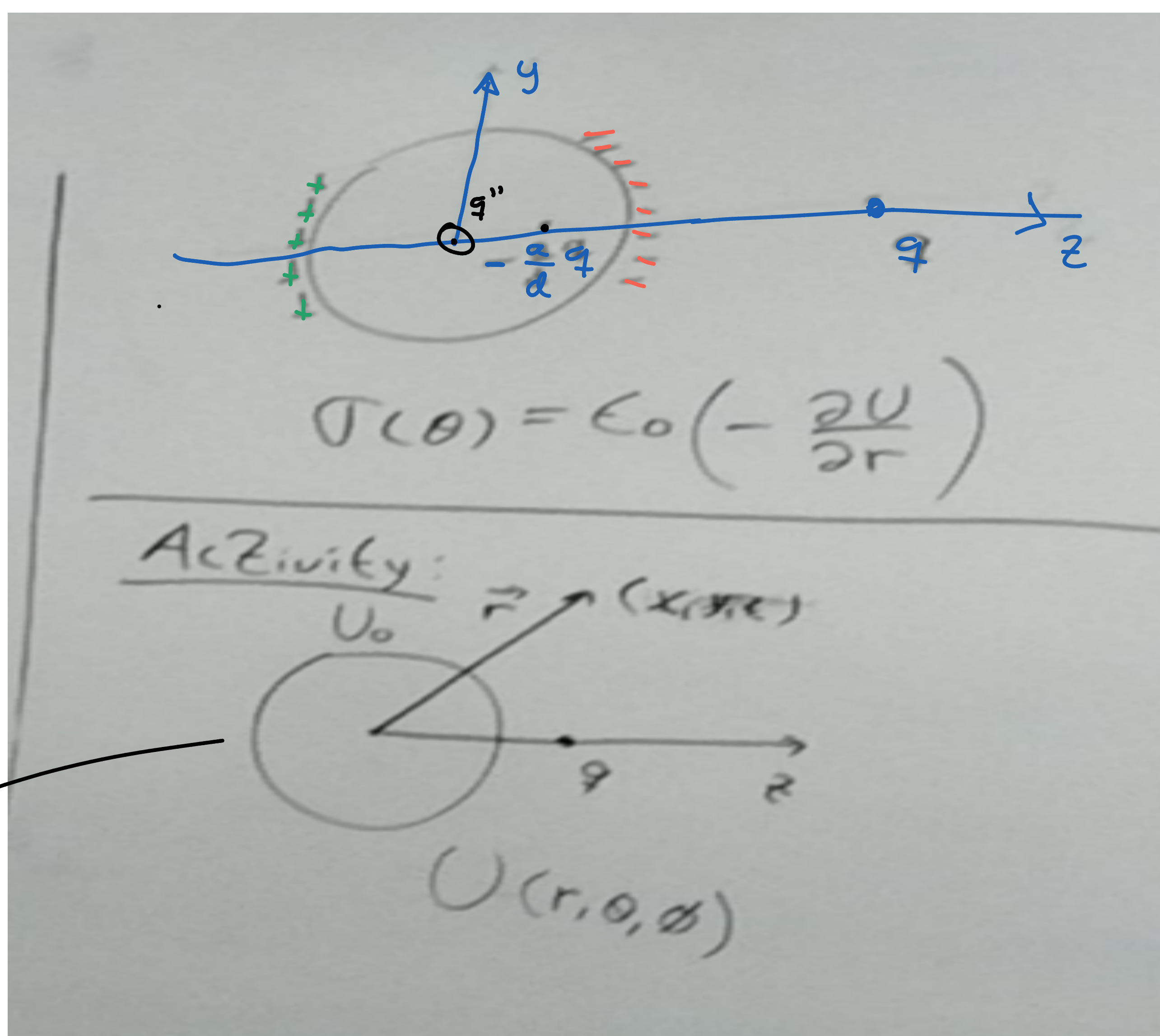
$$V(r, \theta, \phi) \Big|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{r^2 + d^2 - 2rd\cos\theta}} + \frac{q'}{\sqrt{r^2 + b^2 - 2rb\cos\theta}} \right] = 0$$

Temas

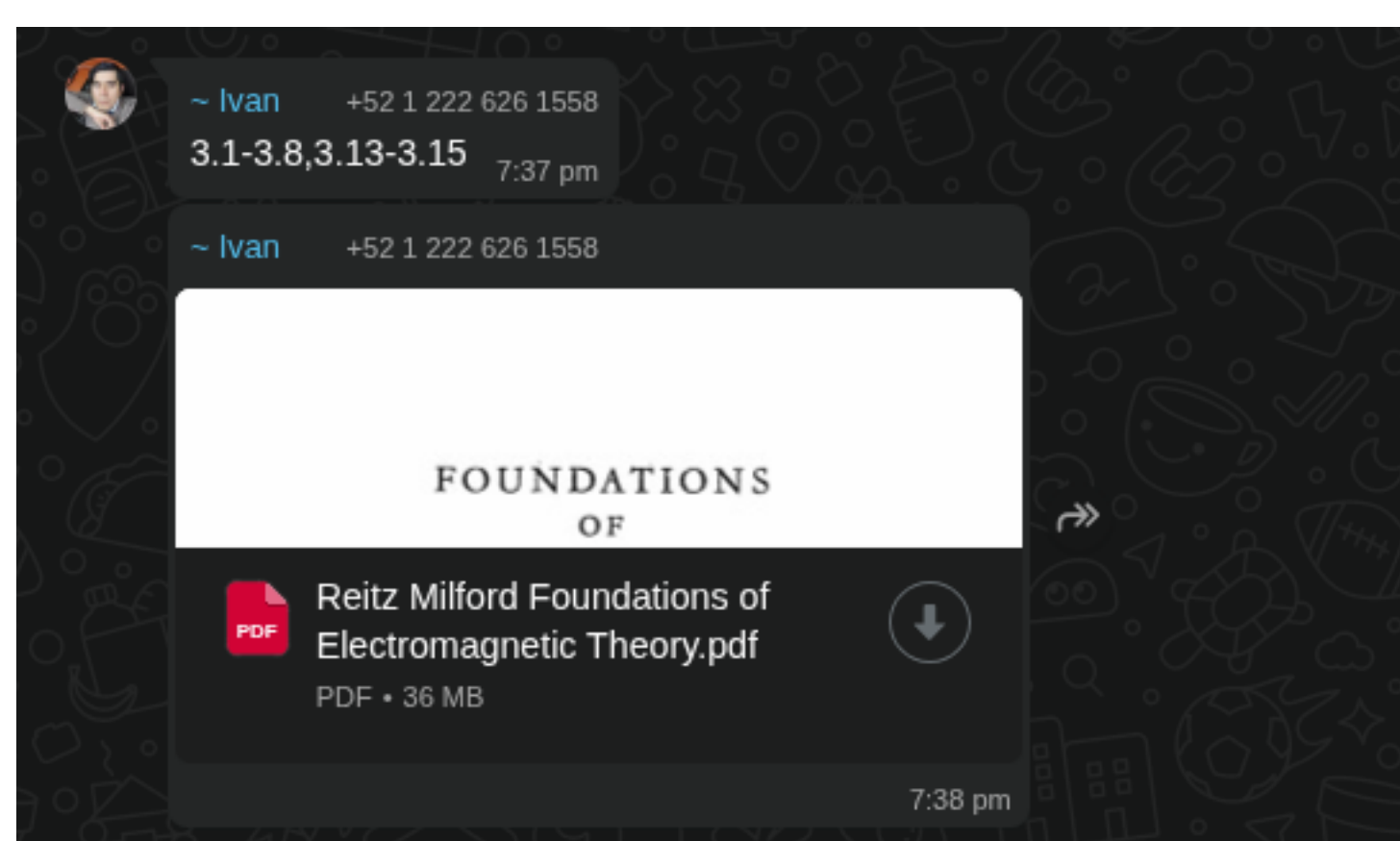
$$V|_{r=a} = \frac{1}{4\pi\epsilon_0} \left[\frac{q \left(\frac{a}{d}\right)}{\sqrt{a^2 + d^2 - 2ad\cos\theta}} + \frac{q'}{\sqrt{a^2 + b^2 - 2ab\cos\theta}} \right] = 0$$

$$\Rightarrow q \left(\frac{a}{d}\right) + q' = 0$$

$$q' = -\frac{a}{d} q$$



Ejercicios para examen.



$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{d^2 + r^2 - 2dr\cos\theta}} - \frac{q \frac{a}{d}}{\sqrt{b^2 + r^2 - 2rb\cos\theta}} + \frac{q''}{r} \right]$$